

EF-P Translation Stall Rescue: A Review-Style Synthesis

*Commissioned review of the *Pseudomonas putida* KT2440 UniPathway UPA00345 module for elongation factor P-dependent rescue of stalled cytosolic ribosomes, centered on *efp* (PP_1858, UniProtKB:Q88LS0).*

1. Executive Summary

Elongation factor P (EF-P) is a universally conserved bacterial translation factor whose single, well-defined job is to rescue 70S ribosomes that stall when the peptidyl-transferase center must join intrinsically poor peptide-bond substrates — most prominently consecutive proline residues (PPP, PPG, and longer strings). EF-P is a **structural mimic of tRNA**: it binds into the space between the ribosomal P and E sites, engages the CCA-end of the P-site peptidyl-tRNA through a post-translationally modified residue at the tip of its first domain, stabilizes the peptidyl-tRNA in a productive geometry, and thereby accelerates Pro–Pro bond formation. This is the entirety of the UPA00345 pathway bucket as it applies to *Pseudomonas putida* KT2440: a **single-gene module**, centered on *efp* (PP_1858 / Q88LS0), performing EF-P-dependent stimulation of peptide-bond formation and stall rescue.

The system's most important boundary condition is that EF-P requires a **lineage-specific post-translational modification** to reach full activity. In enterobacteria (*E. coli*, *Salmonella*) the modification is β -lysylation of Lys34 (via EpmA/EpmB); in pseudomonads and several other lineages (*P. aeruginosa*, *P. putida*, *Shewanella*, *Neisseria*) it is instead **rhamnosylation of Arg32**, catalyzed by the glycosyltransferase EarP using dTDP-L-rhamnose as the sugar donor. For the KT2440 scope defined here, EarP-dependent Arg32 rhamnosylation and the dTDP-L-rhamnose biosynthetic supply are the **activation context** for EF-P — biologically essential for full function, but *not* members of the single-gene UPA00345 pathway. Keeping this distinction crisp is central to the scope of this review.

Three further facts frame the biology. First, EF-P is **ancient and universal**: it is the bacterial ortholog of eukaryotic eIF5A and archaeal aIF5A, sharing an L-shaped, tRNA-like architecture and a conserved modified basic residue at the domain-I tip. Second, EF-P's requirement is

usually **conditional** rather than absolute — cells lacking it grow, albeit with fitness and regulatory defects — but it is essential in some species (e.g., *Neisseria meningitidis*), reflecting how many polyproline-containing essential proteins a given genome carries. Third, polyproline rescue is **partially redundant**: paralogs (EfpL/YeiP) and unrelated factors (YfmR, YebC2, the ABCF ATPase Uup) provide alternative or overlapping routes to resolving proline-induced arrest. EF-P nonetheless remains mechanistically distinct from trans-translation, ribosome recycling, and ribosome-associated quality control (RQC), which act on genuinely dead-end or aberrant complexes rather than on transiently slow but competent elongation.

2. Definition and Biological Boundaries

2.1 What is included

The UPA00345 module comprises a **single molecular player and a single activity**:

Component	Identity	Role
efp	PP_1858 / UniProtKB:Q88LS0 (PSEPK EF-P)	Translation elongation factor that binds stalled 70S ribosomes and stimulates peptide-bond formation at hard-to-translate motifs

The biological function is the **EF-P-dependent rescue of stalled translation elongation** in the cytosol: EF-P recognizes and binds a ribosome whose P-site holds a peptidyl-tRNA that is a poor peptide-bond donor (canonically peptidyl-prolyl-tRNA), stabilizes that tRNA, and promotes formation of the next peptide bond with an incoming (typically prolyl-)aminoacyl-tRNA. This is elongation-phase catalysis assistance, not initiation, termination, or recycling.

2.2 What is activation context, not a pathway member

Full EF-P activity in *P. putida* depends on **Arg32 rhamnosylation by EarP**, which in turn depends on **dTDP-L-rhamnose supply** (the RmlABCD nucleotide-sugar pathway). These are biologically obligatory for full function but are, by the scope of this brief, *activation context* rather than steps of the single-gene UPA00345 module. They are discussed here because no honest account of pseudomonad EF-P can omit them, but they should be catalogued separately (e.g., under rhamnose biosynthesis and protein glycosylation) rather than folded into the stall-rescue bucket.

2.3 Neighboring processes often confused with EF-P rescue

Several ribosome-rescue and quality-control systems share vocabulary ("rescue," "stalling") with EF-P but are mechanistically and functionally distinct and **should be treated separately**:

- **trans-translation (tmRNA–SmpB)** and the alternative rescue factors ArfA/ArfB: these act on ribosomes stalled at the **3' end of a broken or nonstop mRNA** (no A-site codon), tagging the nascent chain for degradation and recycling the ribosome. EF-P, by contrast, acts on **competent, translatable** complexes to keep elongation going.
- **Ribosome-associated quality control (RQC)**: resolves stalled/collided ribosomes and targets aberrant nascent chains for degradation. EF-P prevents the stall in the first place rather than dismantling the product.
- **The ABCF ATPases (Uup, YfmR, EttA, etc.)**: some of these overlap functionally with EF-P at proline-dependent stalls (see §5.3), but they are a distinct protein family using ATP hydrolysis, not tRNA mimicry.
- **eIF5A/aIF5A**: the eukaryotic/archaeal orthologs performing the same catalytic role — homologous, not a separate confusable system, but they use a chemically different modification (hypusine) and should not be assumed identical in regulation.

2.4 Competing definitions in the literature

Historically, EF-P was defined (1975) as a factor that **stimulates the peptidyltransferase reaction in vitro** and was thought to act specifically on **first-peptide-bond formation** during early elongation. The modern definition — established in 2013 — is that EF-P acts **throughout elongation** to alleviate stalling at polyproline and related motifs. The review by Lassak and colleagues traces this shift explicitly ([PMID: 32011712](#)). Thus "competing definitions" are largely historical: the polyproline model has superseded the first-bond model, though EF-P's general peptidyltransferase-stimulating activity remains part of the picture.

3. Mechanistic Overview

3.1 The best current model

The consensus sequence of events for EF-P-mediated rescue is:

Ribosome translating an mRNA encoding a polyproline (or related) motif



Peptidyl-prolyl-tRNA in P site = poor peptide-bond DONOR;
incoming prolyl-tRNA = poor ACCEPTOR → elongation STALLS



EF-P (tRNA-shaped) SCANS for a ribosome with a vacant E site



EF-P binds between P and E sites; domain-I tip (modified
Arg32-rhamnose in *P. putida*) reaches the P-site tRNA CCA end



Peptidyl-tRNA STABILIZED; nascent chain reoriented into a
favorable geometry for catalysis



Pro-Pro PEPTIDE BOND forms; ribosome resumes elongation



EF-P dissociates; cycle repeats at the next problematic motif

3.2 Obligatory, conditional, and accessory steps

- **Obligatory:** (i) presence of a stall-prone motif in the mRNA; (ii) a P-site peptidyl-tRNA whose D-arm EF-P can recognize; (iii) a vacant E site to admit EF-P; (iv) engagement of the peptidyl-tRNA CCA-end by EF-P's domain-I tip; (v) stimulation of peptide-bond formation.
- **Conditional (for full activity, lineage-dependent):** post-translational modification of the domain-I tip residue — β -Lys34 (enterobacteria) or rhamnosyl-Arg32 (pseudomonads and others). Unmodified EF-Ps of the PGKGP subfamily can be active without modification, and specific amino-acid substitutions can convert marginally active unmodified EF-Ps into fully functional ones ([PMID: 38635400](#)). So modification is conditional, not universally obligatory.
- **Accessory/context-dependent:** the strength of a given stall depends on the **sequence context upstream** of the motif — some N-terminal residues (Cys, Thr) suppress stalling while others (Arg, His) promote it ([PMID: 25143529](#)). Redundant factors (EfpL, YfmR, YebC2, Uup) are accessory routes rather than core steps.

3.3 Molecular assemblies carrying out each step

The active assembly is the **stalled 70S ribosome + P-site peptidyl-tRNA + EF-P**. Cryo-EM of stalled 70S complexes with and without EF-P shows that EF-P's modification contacts the CCA-end of the P-site tRNA and enforces an alternative nascent-chain conformation that permits favorable substrate geometry (PMID: 29100052). Kinetically, EF-P **scans** for a free E site and can bind any ribosome with a P-site tRNA regardless of functional state, but is **retained selectively** on complexes that actually need help; the modification increases the association rate roughly two-fold (PMID: 39315709).

4. Major Molecular Players and Active Assemblies

4.1 EF-P — the tRNA mimic (Finding F003)

The crystal structure of *Thermus thermophilus* EF-P revealed **three β -barrel domains arranged in an L-shape**, with 65-Å and 53-Å arms meeting at $\sim 95^\circ$, "reminiscent of tRNA" (PMID: 15210970: "*The three domains of EF-P are arranged in an 'L' shape, with 65- and 53-Å long arms at an angle of 95 degrees, which is reminiscent of tRNA.*"). The tip of domain I carries a conserved positively charged residue — the modification site. The same study established that Eukarya and Archaea possess the homolog **eIF5A/aIF5A** ("*Eukarya and Archaea have an EF-P homologue, eukaryotic initiation factor 5A (eIF-5A).*"), marking EF-P as part of a deeply conserved, cross-domain protein family. This structural mimicry explains how EF-P slots into the tRNA-binding channel of the ribosome between the P and E sites.

4.2 The catalytic action on the ribosome (Finding F001)

EF-P performs two coupled actions: it **stabilizes the P-site peptidyl-tRNA** in the catalytic center and **promotes peptide-bond formation**. Doerfel et al. (2013) demonstrated that EF-P prevents stalling at PPG/PPP and longer proline strings and accelerates peptide-bond formation (PMID: 23239624: "*EF-P promotes peptide-bond formation and stabilizes the peptidyl-transfer RNA in the catalytic center of the ribosome.*"). Huter et al. (2017) provided the structural basis: "*Binding of EF-P stabilizes the P-site tRNA, particularly via interactions between its modification and the CCA end, thereby enforcing an alternative conformation of the polyproline-containing nascent chain, which allows a favorable substrate geometry for peptide bond formation*" (PMID: 29100052).

4.3 The activation module — EarP, Arg32, and dTDP-L-rhamnose (Finding F002)

In pseudomonads, EF-P is activated by a modification chemically distinct from the enterobacterial one. Rajkovic et al. (2015) showed that *P. aeruginosa* EF-P carries a **single cyclic rhamnose on Arg32**, at the position equivalent to *E. coli* β -Lys34, and that the enzyme **EarP uses dTDP-L-rhamnose in vitro** (PMID: 26060278: "*Structural analyses of P. aeruginosa EF-P revealed the attachment of a single cyclic rhamnose moiety to an Arg residue at a position equivalent to that at which β -Lys is attached to E. coli EF-P.*" and "*In vitro assays confirmed the ability of EarP to use dTDP-L-rhamnose as a substrate for the posttranslational glycosylation of EF-P.*"). Loss of *efp*, *earP*, or *rmlC* (rhamnose biosynthesis) abolishes Pro4-GFP reporter output. Krafczyk et al. (2017) solved the EarP structure and confirmed Arg32 as the acceptor (PMID: 28951478: "*EF-P is rhamnosylated on arginine 32 by the glycosyltransferase EarP.*"). Per the review scope, **this activation module is context, not a member of the single-gene UPA00345 pathway** — but it is functionally indispensable in *P. putida*.

4.4 Substrate recognition and kinetics (Finding F004)

EF-P's selectivity derives partly from the **tRNA it recognizes** and partly from its **binding kinetics**. Katoh et al. (2016) identified the **9-nucleotide D-loop closed by a stable D-stem** of tRNA(Pro) — a feature shared with tRNA(fMet) — as the crucial recognition determinant (PMID: 27216360: "*We show that the 9-nt D-loop closed by the stable D-stem sequence in tRNA(Pro) is a crucial recognition determinant for EF-P.*"). Mudryi et al. (2024), using FRET kinetics, showed that EF-P **rapidly scans for a free E site and can bind any ribosome containing a P-site tRNA regardless of functional state** (PMID: 39315709: "*EF-P rapidly scans for a free E site and can bind to any ribosome containing a P-site tRNA, regardless of the ribosome's functional state.*"), being retained only on complexes that need rescue — a scanning-engagement mechanism in which the modification tunes affinity.

4.5 Redundant and alternative factors (Finding F005)

EF-P is not the only route to resolving proline-induced arrest:

Factor	Family	Evidence
EfpL / YeiP	EF-P paralog	<i>"numerous bacteria possess an EF-P paralog called EfpL (also known as YeiP) of unknown function"; resolves overlapping and additional arrest motifs (PMID: 39622818)</i>
YebC2	Transcription-factor-like	<i>"YebC2 can reduce ribosome stalling and support cellular fitness in the absence of EF-P and YfmR" (PMID: 40215226)</i>
Uup	ABCF ATPase	<i>"Uup for poly-proline-dependent stalling" within a division of labor across ABCFs (PMID: 38661232)</i>
YfmR	ABCF ATPase	Acts alongside EF-P; its loss with EF-P is compensated by YebC2 (PMID: 40215226)

This redundancy is one reason EF-P's requirement is usually conditional.

4.6 Physiological output — tuning the polyproline proteome (Finding F006)

Ude et al. (2013), co-discoverers of the polyproline function, showed that *"In the absence of EF-P, ribosomes stall at polyproline stretches, whereas the presence of EF-P alleviates the translational stalling"* and demonstrated *"the physiological relevance of EF-P to fine-tune the expression of the polyproline-containing pH receptor CadC to levels necessary for an appropriate stress response"* (PMID: 23239623). Because bacterial, archaeal, and eukaryotic proteomes contain hundreds to thousands of polyproline proteins, EF-P is best understood as a **proteome-wide tuner** of a specific, functionally enriched subset (transporters, regulators, enzymes with proline-rich motifs), not merely a rescuer of rare catastrophic stalls.

5. Evolutionary and Cell-Biological Variation

5.1 Cross-domain conservation and origin (Finding F003)

EF-P is **ancient and universal**. Its L-shaped, tRNA-mimicking architecture and its cross-domain orthology to eIF5A (eukaryotes) and aIF5A (archaea) place its origin at or before the last universal common ancestor of the translation apparatus (PMID: 15210970). The tRNA-like fold and the conserved modified basic tip are the deeply conserved core; the *chemistry* of the modification is the variable, lineage-specific elaboration.

5.2 Variation in the modification chemistry

The single conserved job (activate the domain-I tip) is met by at least three distinct solutions:

Lineage	Modified residue	Modification	Enzymes
Enterobacteria (<i>E. coli</i> , <i>Salmonella</i>)	Lys34	(R)- β -lysylation	EpmA (YjeA), EpmB (YjeK), EpmC
Pseudomonads, <i>Shewanella</i> , <i>Neisseria</i>	Arg32	Cyclic rhamnosylation	EarP + dTDP-L-rhamnose (RmlABCD)
Eukaryotes / Archaea (eIF5A/aIF5A)	Lys	Hypusination	Deoxyhypusine synthase, deoxyhypusine hydroxylase
PGKGP-subfamily bacteria	—	None (or context-compensated)	— (PMID: 38635400)

The β -lysyl pathway is conserved in only ~26–28% of bacteria; the rhamnosyl pathway serves a substantial additional fraction, and some EF-Ps require no modification at all. This is a striking case of **convergent activation of a conserved core** by non-homologous chemistries.

5.3 Redundancy and alternative routes across lineages

As catalogued in §4.5, some bacteria carry EF-P paralogs (EfpL/YeiP) and unrelated factors (YebC2, ABCF Uup/YfmR) that partially substitute for or complement EF-P (PMID: 39622818; PMID: 40215226; PMID: 38661232). The distribution of these backups varies across lineages and helps explain why the *efp* deletion phenotype ranges from mild to lethal.

5.4 Conditional vs. essential requirement

In *E. coli*, *Salmonella*, *P. aeruginosa*, and *Shewanella*, EF-P and its modification are **dispensable for viability** (though important for fitness and virulence). In *Neisseria meningitidis*, by contrast, EF-P and its active-site Arg are **essential for cell viability** (PMID: 26840407). The likely explanation is that *N. meningitidis* has more essential polyproline-containing proteins and/or a lower basal ability to translate proline stretches without EF-P. This makes EarP a candidate antibacterial target in that pathogen.

5.5 The mitochondrial and eukaryotic parallels

Eukaryotic cytosolic translation uses eIF5A for the identical role. Notably, **mitochondria lack an obvious EF-P/eIF5A ortholog**, and recent work implicates **TACO1** as the functional counterpart that alleviates mitoribosome stalling at polyproline stretches in human mitochondria (PMID: 39036954). This is an example of an **analogous, non-homologous solution** evolving to meet the same constraint in a different compartment — and it underscores that one should not assume the bacterial cytosolic mechanism generalizes to all compartments.

5.6 Horizontal transfer and proteome co-evolution

The *efp* gene has been **horizontally transferred** multiple times among bacteria, events associated with the loss of highly conserved polyproline motifs and even of entire polyproline-containing proteins (e.g., the proteases Lon, ClpC, FtsH and the tRNA synthetases ValS, IleS1, IleS2) (PMID: 39189989). This co-evolution shows that the EF-P system and the polyproline proteome are **coupled**: perturbing EF-P leaves lasting genomic traces in the distribution of proline motifs, particularly near ATP-binding regions.

6. Constraints, Dependencies, and Failure Modes

6.1 Ordering constraints

- EF-P acts **only after** a peptidyl-tRNA occupies the P site and **only when the E site is vacant** — it cannot bind an E-site-occupied ribosome (PMID: 39315709). Thus its action is temporally gated within each elongation cycle.
- EF-P engages **before** peptide-bond formation, stabilizing the substrate; it is not a post-catalysis factor.
- Modification (where required) must occur **before** the ribosome-binding step: unmodified EF-P in modification-dependent lineages is marginally active.

6.2 Compartment and substrate specificity

- **Cytosolic 70S ribosomes only** — this is the scope of UPA00345. Organellar (mitochondrial) stalling is handled by other factors (e.g., TACO1; PMID: 39036954).

- **Substrate-specific:** EF-P is recruited via recognition of the P-site tRNA D-arm, with tRNA(Pro) (and tRNA(fMet)) sharing the key 9-nt D-loop/stable D-stem determinant ([PMID: 27216360](#)). The stalling that requires rescue is concentrated at proline-proline and proline-rich motifs, extended by problematic flanking residues and specific upstream contexts ([PMID: 25143529](#)).

6.3 Failure modes

- **Loss of EF-P** (Δefp): ribosome stalling at polyproline motifs, reduced/altered expression of polyproline proteins, fitness and stress-response defects (e.g., impaired CadC-mediated acid response; [PMID: 23239623](#)), attenuated virulence, and — in some species — lethality ([PMID: 26840407](#)).
- **Loss of the modification machinery** ($\Delta earP$, $\Delta rmlC$ in pseudomonads; $\Delta epmA/B$ in enterobacteria): phenocopies partial loss of EF-P because the factor is present but under-active ([PMID: 26060278](#)).
- **Compensation:** partial rescue by EfpL, YebC2, YfmR, and Uup mitigates these failures to varying degrees ([PMID: 39622818](#); [PMID: 40215226](#); [PMID: 38661232](#)).

6.4 Evidence ruling out otherwise-plausible paths

- EF-P is **not** a first-peptide-bond-only factor: the 2013 studies and subsequent work show it acts throughout elongation ([PMID: 32011712](#); [PMID: 23239624](#)).
- EF-P is **not** a nonstop/no-A-site rescue factor — that niche belongs to tmRNA-SmpB/ArfA/ArfB. EF-P rescues **competent, translatable** complexes.
- Modification is **not strictly obligatory in all lineages** — PGKGP-subfamily EF-Ps function unmodified, ruling out a universal "modification-is-essential" model ([PMID: 38635400](#)).

7. Controversies and Open Questions

Strongly supported claims. The core mechanism — EF-P is a tRNA-mimic that binds the vacant E site of a stalled ribosome, stabilizes the P-site peptidyl-tRNA via a modified domain-I tip, and stimulates polyproline peptide-bond formation — is supported by convergent biochemical (Doerfel/Ude 2013), structural (cryo-EM 2017), and kinetic (FRET 2024) evidence

([PMID: 23239624](#); [PMID: 23239623](#); [PMID: 29100052](#); [PMID: 39315709](#)). The existence of the alternative pseudomonad Arg32-rhamnosyl modification is likewise firmly established ([PMID: 26060278](#); [PMID: 28951478](#)).

Areas of disagreement or indirect evidence.

1. **Role of the modification.** The classic view held modification to be strictly required for activity; newer work shows a more nuanced, sometimes dispensable role, with unmodified EF-Ps active in certain sequence contexts ([PMID: 32011712](#); [PMID: 38635400](#)). How much of the modification's contribution is affinity (the ~2-fold association effect; [PMID: 39315709](#)) versus catalytic geometry remains only partly resolved.
2. **Full motif spectrum.** Beyond canonical PPP/PPG, the complete set of arrest-inducing motifs — and how upstream sequence context modulates them — is still being mapped, with reporter and profiling approaches sometimes disagreeing ([PMID: 25143529](#); [PMID: 39425678](#)).
3. **Division of labor among rescue factors.** How EF-P, EfpL, YebC2, YfmR, and Uup partition the polyproline/hard-to-translate motif space — and whether this partition is conserved across lineages — is an active question ([PMID: 39622818](#); [PMID: 40215226](#); [PMID: 38661232](#)).
4. **Cross-organism extrapolation.** Much mechanistic detail comes from *E. coli* and *T. thermophilus*; the pseudomonad system is inferred largely by homology plus the specific rhamnosylation data. The essential-vs-conditional distinction (*Neisseria* vs. *E. coli*) warns against assuming one organism's phenotype generalizes ([PMID: 26840407](#)).
5. **Compartmental non-homology.** The mitochondrial solution (TACO1) is non-homologous, cautioning against assuming the bacterial mechanism is the only way to meet the polyproline constraint ([PMID: 39036954](#)).

Most important open questions. (i) What is the precise catalytic contribution of the rhamnosyl modification at the atomic level in *P. putida* EF-P? (ii) What is the full, context-aware motif spectrum that EF-P rescues in pseudomonads specifically? (iii) How do the redundant factors distribute the load in vivo, and does EarP represent a viable narrow-spectrum antibacterial target in pathogens where EF-P is essential?

8. Key References

PMID	Contribution	Relevance
23239624	Doerfel et al. 2013 — EF-P stabilizes peptidyl-tRNA and promotes peptide-bond formation at proline runs	Core mechanism (F001)
23239623	Ude et al. 2013 — EF-P alleviates polyproline stalling; tunes CadC	Co-discovery, physiology (F006)
29100052	Huter et al. 2017 — cryo-EM structural basis of stall and rescue	Structural mechanism (F001)
15210970	Hanawa-Suetsugu et al. 2004 — <i>T. thermophilus</i> EF-P structure; tRNA mimicry; eIF5A orthology	Conservation/origin (F003)
26060278	Rajkovic et al. 2015 — Arg32 rhamnosylation via EarP/dTDP-L-rhamnose in <i>P. aeruginosa</i>	Pseudomonad activation (F002)
28951478	Krafczyk et al. 2017 — EarP structure; Arg32 acceptor	Pseudomonad activation (F002)
27216360	Katoh et al. 2016 — tRNA(Pro) D-arm recognition determinant	Substrate specificity (F004)
39315709	Mudryi et al. 2024 — scanning-engagement kinetics	Recruitment mechanism (F004)
39622818	Sieber et al. 2024 — EfpL/YeiP paralog	Redundancy (F005)
40215226	Hong et al. 2025 — YebC2 rescues $\Delta\text{efp } \Delta\text{yfmr}$	Redundancy (F005)
38661232	Chadani et al. 2024 — ABCF Uup handles poly-proline stalling	Redundancy (F005)
32011712	Lassak review — history and evolving definition of EF-P	Definitions/controversy
38635400	Unmodified EF-P design principles (PGKGP subfamily)	Modification conditionality
25143529	Upstream sequence context modulates stalling	Motif determinants
26840407	EF-P essential in <i>N. meningitidis</i>	Conditional vs. essential
39036954	TACO1 as mitochondrial polyproline-rescue factor	Compartmental non-homology

PMID	Contribution	Relevance
39189989	Horizontal transfer of <i>efp</i> and polyproline proteome co-evolution	Evolution
25364902	Proline-repeat proteome and eIF5A across kingdoms	Proteome scope
39425678	Dual reporter to map EF-P-alleviated pausing motifs	Motif mapping

Appendix A: Limitations and Knowledge Gaps of This Review

- **Data source:** This synthesis is literature-based; no primary sequence, structural, or omics dataset was analyzed de novo for *P. putida* KT2440 specifically. Pseudomonad-specific mechanistic claims rest on *P. aeruginosa*/*Shewanella*/*Neisseria* data plus homology to KT2440 EF-P (Q88LS0).
- **Organism mixing:** Mechanistic detail derives disproportionately from *E. coli* and *T. thermophilus*. Extrapolation to *P. putida* is reasonable but not experimentally confirmed at every step here.
- **Modification chemistry:** The atomic-level catalytic role of the rhamnosyl modification (vs. the β -lysyl modification) is incompletely resolved.
- **Redundancy quantification:** The in vivo load-sharing among EF-P, EfpL, YebC2, YfmR, and Up is not quantitatively established for pseudomonads.

Appendix B: Proposed Follow-up Experiments / Actions

1. **KT2440-specific reporter panel.** Deploy a dual-reporter polyproline pausing assay ([PMID: 39425678](#)) in *P. putida* KT2440 across Δefp , $\Delta earP$, and $\Delta rmlC$ backgrounds to define the organism-specific motif spectrum and quantify the rhamnosylation dependence directly.
2. **Ribosome profiling** of wild-type vs. Δefp KT2440 to map genome-wide stall sites and identify which polyproline-containing proteins (transporters, regulators, proteases) are EF-P-dependent in this organism.
3. **Structural determination** of rhamnosyl-EF-P (Q88LS0) on a stalled *P. putida* 70S complex by cryo-EM to resolve the catalytic contribution of the Arg32-rhamnose modification.

4. **Redundancy dissection.** Systematically delete EF-P backup candidates (EfpL, YebC2, ABCF ATPases) singly and in combination with Δefp to measure epistasis and quantify load-sharing.
 5. **EarP as antibacterial target.** Assess whether EarP inhibition phenocopies EF-P loss and screen for EarP inhibitors, prioritizing pathogens where EF-P is essential ([PMID: 26840407](#)).
 6. **Comparative proteomics** to test the horizontal-transfer/proteome-coupling hypothesis ([PMID: 39189989](#)) within the pseudomonad lineage.
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Prepared as a commissioned review synthesis. Scope centered on the single-gene UPA00345 module (efp / PP_1858 / Q88LS0); EarP-dependent Arg32 rhamnosylation and dTDP-L-rhamnose supply treated as activation context, not pathway members.

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